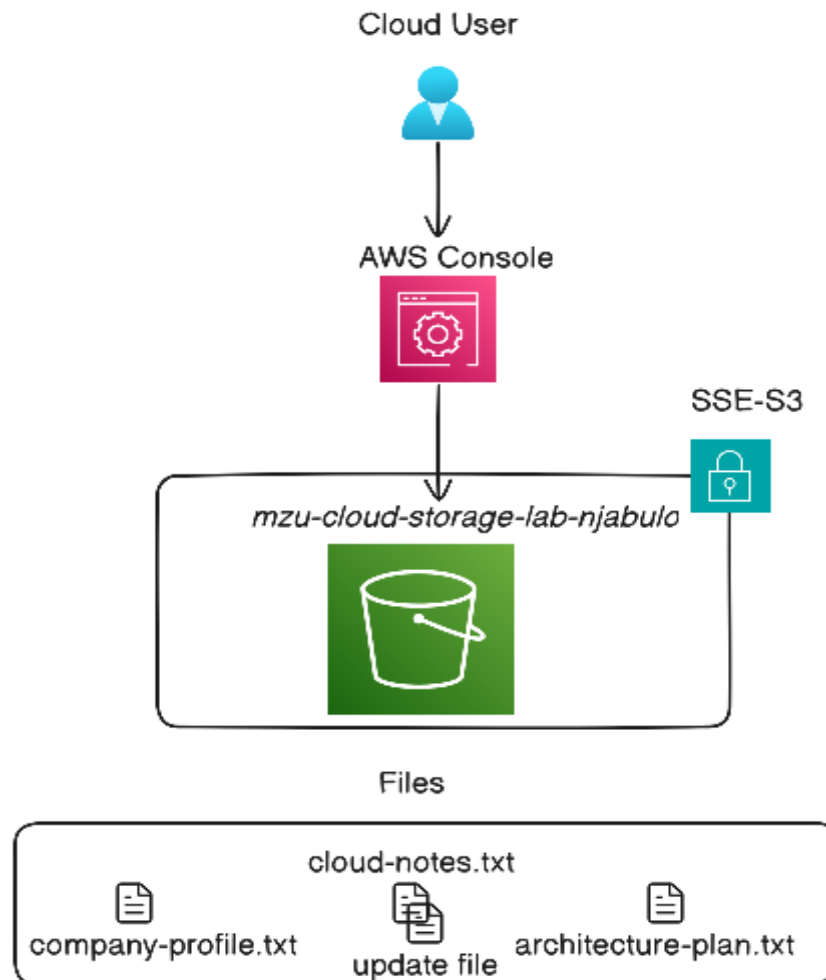


Cloud Storage (S3) & Local Version Control (Git)

LAB 1 – CREATE YOUR FIRST AMAZON S3 BUCKET



Task 1 Create Bucket S3

Name

- *mzu-cloud-storage-lab-njabulo*

AWS Region

- *US East (N. Virginia) us-east-1*

Name	AWS Region	Creation date
☒ mzu-cloud-storage-lab-njabulo	US East (N. Virginia) us-east-1	July 30, 2026, 15:01:36 (UTC+02:00)
☐ mzuca-shared-responsibility-lot42	US East (N. Virginia) us-east-1	July 21, 2026, 12:52:18 (UTC+02:00)

Block Access

- Enable
- Reason Business data should not be publicly accessible

Block public access (bucket settings) Edit

Public access is granted to buckets and objects through access control lists (ACLs), bucket policies, access point policies, or all. In order to ensure that public access to all your S3 buckets and objects is blocked, turn on Block all public access. These settings apply only to this bucket and its access points. AWS recommends that you turn on Block all public access, but before applying any of these settings, ensure that your applications will work correctly without public access. If you require some level of public access to your buckets or objects within, you can customize the individual settings below to suit your specific storage use cases. [Learn more](#)

Block all public access
On

▼ Individual Block Public Access settings for this bucket

- ☒ **Block public access to buckets and objects granted through new access control lists (ACLs)**
S3 will block public access permissions applied to newly added buckets or objects, and prevent the creation of new public access ACLs for existing buckets and objects. This setting doesn't change any existing permissions that allow public access to S3 resources using ACLs.
- ☒ **Block public access to buckets and objects granted through any access control lists (ACLs)**
S3 will ignore all ACLs that grant public access to buckets and objects.
- ☒ **Block public access to buckets and objects granted through new public bucket or access point policies**
S3 will block new bucket and access point policies that grant public access to buckets and objects. This setting doesn't change any existing policies that allow public access to S3 resources.
- ☒ **Block public and cross-account access to buckets and objects through any public bucket or access point policies**
S3 will ignore public and cross-account access for buckets or access points with policies that grant public access to buckets and objects.

Bucket Versioning

- Enable
- Reason: Allows recovery of previous object versions

Bucket Versioning Edit

Versioning is a means of keeping multiple variants of an object in the same bucket. You can use versioning to preserve, retrieve, and restore every version of every object stored in your Amazon S3 bucket. With versioning, you can easily recover from both unintended user actions and application failures. [Learn more](#)

Bucket Versioning
Enabled

Multi-factor authentication (MFA) delete
An additional layer of security that requires multi-factor authentication for changing Bucket Versioning settings and permanently deleting object versions. To modify MFA delete settings, use the AWS CLI, AWS SDK, or the Amazon S3 REST API. [Learn more](#)
Disabled

Encryption

- Enable: Server-side encryption
- Choose: SSE-S3

Default encryption Edit

Server-side encryption is automatically applied to new objects stored in this bucket.

Encryption type [Info](#)
Server-side encryption with Amazon S3 managed keys (SSE-S3)

Bucket Key
When KMS encryption is used to encrypt new objects in this bucket, the bucket key reduces encryption costs by lowering calls to AWS KMS. [Learn more](#)
Enabled

Blocked encryption types [Info](#)
Server-Side Encryption with Customer-Provided Keys (SSE-C)

Upload & Manage Objects

Task1

Create

- *company-profile.txt*
- *cloud-notes.txt*
- *architecture-plan.txt*



architecture-plan



cloud-notes



company-profile

Task 2

Objects (3)

Copy S3 URICopy URLLDownloadOpen inNewDeleteActionsCreate folderUpload

Find objects by prefixShow versions

☐	Name	Type	Last modified	Size	Storage class
☐	architecture-plan.txt	txt	July 30, 2026, 15:27:59 (UTC+02:00)	0 B	Standard
☐	cloud-notes.txt	txt	July 30, 2026, 15:28:00 (UTC+02:00)	0 B	Standard
☐	company-profile.txt	txt	July 30, 2026, 15:28:00 (UTC+02:00)	0 B	Standard

Upload

Task 3

Explore Object Information

Example

- cloud-notes.txt

cloud-notes.txtInfo

Copy S3 URIDownloadOpen inNew

PropertiesPermissionsVersions

Object overview

Owner

03dd4afa57a2056da3ba98b8f5d1b21c8ecbab1a69a444f94edb7107676e8bbe

AWS Region

US East (N. Virginia) us-east-1

Last modified

July 30, 2026, 15:28:00 (UTC+02:00)

Size

0 B

Type

txt

Key

cloud-notes.txt

S3 URI

s3://mzu-cloud-storage-lab-njabulo/cloud-notes.txt

Amazon Resource Name (ARN)

arn:aws:s3::mzu-cloud-storage-lab-njabulo/cloud-notes.txt

Entity tag (Etag)

d41d8cd98f00b204e9800998ecf8427e

Object URL

https://mzu-cloud-storage-lab-njabulo.s3.us-east-1.amazonaws.com/cloud-notes.txt

Storage Class

Storage class

Amazon S3 offers a range of storage classes designed for different use cases. [Learn more](#) or see [Amazon S3 pricing](#)

Storage class

Standard

Edit

Encryption Status

Server-side encryption settingsInfo

Server-side encryption protects data at rest.

Encryption typeInfo

Server-side encryption with Amazon S3 managed keys (SSE-S3)

Edit

Lab 3 - TEST S3 VERSIONING

Task 1

Modify a file

- cloud-notes.txt
- blank notes

☐	Name	Type	Last modified	Size	Storage class
☐	architecture-plan.txt	txt	July 30, 2026, 15:27:59 (UTC+02:00)	0 B	Standard
☒	cloud-notes.txt	txt	July 30, 2026, 15:28:00 (UTC+02:00)	0 B	Standard
☐	company-profile.txt	txt	July 30, 2026, 15:28:00 (UTC+02:00)	0 B	Standard

Updated

- AWS Cloud Advanced Notes



cloud-notes - Notepad

File Edit Format View Help

AWS Cloud Storage Advanced Notes

Task 2

Upload Updated File

Files and folders

Configuration

Files and folders (1 total, 34.0 B)

Name

cloud-notes.txt

Folder

-

Type

text/plain

Size

34.0 B

Status

Succeeded

Error

-

Objects

Metadata

Properties

Permissions

Metrics

Management

File systems - new

Access Points

Objects (4)

Copy S3 URI

Copy URL

Download

Open i²

Delete

Actions

Create folder

Upload

Find objects by prefix

Show versions

☐	Name	Type	Version ID	Last modified	Size	Storage class
☐	architecture-plan.txt	txt	48J_sH4mwsf6DL_a55QRKW9KbG UTPF6	July 30, 2026, 15:27:59 (UTC+02:00)	0 B	Standard
☐	cloud-notes.txt	txt	DTY4U_Xnh5nB0ypCuazLIQu2Yn8 6e9M	July 30, 2026, 18:21:59 (UTC+02:00)	34.0 B	Standard
☐	cloud-notes.txt	txt	buxKFuPNW9VPP01rLVOG2_uWgw 7WY3fh	July 30, 2026, 15:28:00 (UTC+02:00)	0 B	Standard
☐	company-profile.txt	txt	Q_UckbR0ulZ37smV9xKF_a6ysK9r lhf	July 30, 2026, 15:28:00 (UTC+02:00)	0 B	Standard

Task 3

View Versions

cloud-notes.txt [Info](#) [Copy S3 URI](#) [Download](#) [Open i²](#) [Object actions](#)

Version ID: [buxKFuPNW9VPP01rLV0G2.uWgw7WY3fh](#)

[Properties](#) [Permissions](#) [Versions](#)

Versions (2) [Download](#) [Open i²](#) [Delete](#) [Actions](#)

☐	Version ID	Type	Last modified	Size	Storage class
☐	DTY4U_Xnh5n80ypCuaxLlUQu2Yn866... (Current version)	txt	July 30, 2026, 18:21:59 (UTC+02:00)	34.0 B	Standard
☐	buxKFuPNW9VPP01rLV0G2.uWgw...	txt	July 30, 2026, 15:28:00 (UTC+02:00)	0 B	Standard

LAB 4 - INTRODUCTION TO GIT

Scenario

You are a cloud engineer managing infrastructure code.

You need a system to track changes.

Prerequisites

Install:

Git

Verify:

- ***git --version***

Expected:

git version **2.55.0**

```
C:\Users\njabulo>git --version
git version 2.55.0.windows.3

C:\Users\njabulo>
```

TASK 1 — Create Project Folder

Create:

cloud-storage-project

- ***mkdir cloud-storage-project***

Navigate:

- ***cd cloud-storage-project***

```
C:\Users\njabulo>cd PROJECTS  
C:\Users\njabulo\PROJECTS>cd cloud-storage-project  
C:\Users\njabulo\PROJECTS\cloud-storage-project>
```

TASK 2 — Initialize Git Repository

Run:

git init

Expected:

Initialized empty Git repository

```
C:\Users\njabulo\PROJECTS\cloud-storage-project>git init  
Reinitialized existing Git repository in C:/Users/njabulo/PROJECTS/cloud-storage-project/.git/  
C:\Users\njabulo\PROJECTS\cloud-storage-project>
```

TASK 3 — Create Documentation File

Create:

s3-architecture.md

Add:

Amazon S3 Architecture

S3 is an object storage service.

```
C:\Users\njabulo\PROJECTS\cloud-storage-project>echo "# Amazon S3 Architecture S3 is an object storage service."> s3-architecture.md  
C:\Users\njabulo\PROJECTS\cloud-storage-project>
```

TASK 4 — Check Status

Run:

- ***git status***

Observe:

Git detects your new file.

```
C:\Users\njabulo\PROJECTS\cloud-storage-project>git status
On branch master

No commits yet

Untracked files:
  (use "git add <file>..." to include in what will be committed)
        s3-architecture.md

nothing added to commit but untracked files present (use "git add" to track)
C:\Users\njabulo\PROJECTS\cloud-storage-project>
```

LAB 5 — COMMIT YOUR FIRST CLOUD PROJECT

TASK 1 — Add File

Command:

- ***git add s3-architecture.md***
- ***git status***

```
C:\Users\njabulo\PROJECTS\cloud-storage-project>
C:\Users\njabulo\PROJECTS\cloud-storage-project>git add s3-architecture.md

C:\Users\njabulo\PROJECTS\cloud-storage-project>git status
On branch master

No commits yet

Changes to be committed:
  (use "git rm --cached <file>..." to unstage)
        new file:   s3-architecture.md

C:\Users\njabulo\PROJECTS\cloud-storage-project>
```

TASK 2 — Commit Changes

Command:

- ***git commit -m "Added S3 architecture documentation"***

```
C:\Users\njabulo\PROJECTS\cloud-storage-project>git commit -m "Added S3 architecture documentation"
[master (root-commit) 8650903] Added S3 architecture documentation
Committer: Njabulo Khumalo <Njabulo.Khumalo@seacom.com>
Your name and email address were configured automatically based
on your username and hostname. Please check that they are accurate.
You can suppress this message by setting them explicitly:

    git config --global user.name "Your Name"
    git config --global user.email you@example.com

After doing this, you may fix the identity used for this commit with:

    git commit --amend --reset-author

1 file changed, 1 insertion(+)
create mode 100644 s3-architecture.md

C:\Users\njabulo\PROJECTS\cloud-storage-project>
```

TASK 3 — View History

Command:

- ***git log***

```
C:\Users\njabulo\PROJECTS\cloud-storage-project>
C:\Users\njabulo\PROJECTS\cloud-storage-project>git log
commit 86509034f62846a74802c5b96a289c14d2438394 (HEAD -> master)
Author: Njabulo Khumalo <Njabulo.Khumalo@seacom.com>
Date: Thu Aug 6 22:54:20 2026 +0200

    Added S3 architecture documentation

C:\Users\njabulo\PROJECTS\cloud-storage-project>
```

Learning Outcome

You have created your first tracked cloud engineering project.

LAB 6 — CREATE CLOUD

ENGINEERING REPOSITORY

Scenario

Your team wants documentation stored centrally.

You will create a GitHub repository.

TASK 1

Create repository:

Example name:

- **mzuca-cloud-storage-lab-bhakede**

TASK 2

Connect Local Repository

Command:

- `git remote add origin YOUR_REPOSITORY_URL`
- `git remote add origin https://github.com/iamnjebs/mzuca-cloud-storage-lab-bhakede.git`

```
C:\Users\njabulo\PROJECTS\cloud-storage-project>
C:\Users\njabulo\PROJECTS\cloud-storage-project>git remote add origin https://github.com/iamnjebs/mzuca-cloud-storage-lab-bhakede.git

C:\Users\njabulo\PROJECTS\cloud-storage-project>git remote -v
origin  https://github.com/iamnjebs/mzuca-cloud-storage-lab-bhakede.git (fetch)
origin  https://github.com/iamnjebs/mzuca-cloud-storage-lab-bhakede.git (push)

C:\Users\njabulo\PROJECTS\cloud-storage-project>
```

TASK 3

Push Code

Commands:

`git branch -M main`

`git push -u origin main`

```
C:\Users\njabulo\PROJECTS\cloud-storage-project>
C:\Users\njabulo\PROJECTS\cloud-storage-project>git branch -M main

C:\Users\njabulo\PROJECTS\cloud-storage-project>git push -u origin main
info: please complete authentication in your browser...
Enumerating objects: 3, done.
Counting objects: 100% (3/3), done.
Delta compression using up to 4 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (3/3), 300 bytes | 100.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/iamnjebs/mzuca-cloud-storage-lab-bhakede.git
 * [new branch]      main -> main
branch 'main' set up to track 'origin/main'.

C:\Users\njabulo\PROJECTS\cloud-storage-project>git branch -M main

C:\Users\njabulo\PROJECTS\cloud-storage-project>git push -u origin main
branch 'main' set up to track 'origin/main'.
Everything up-to-date

C:\Users\njabulo\PROJECTS\cloud-storage-project>
```

LAB CHECKPOINT

Confirm:

- ☐ Repository created
- ☐ Code pushed successfully

☐ Files visible online